Trading System Analysis

Holding Duration & Loss Risk

Executive Summary

This report quantifies the relationship between trade holding duration and loss probability across two automated forex trading strategies on the MQL5 platform. The analysis covers 2,423,067 cleaned trades spanning multiple instruments and signal providers.

Headline finding

Loss probability rises monotonically with holding duration: from 16.1% for trades held under one hour, to 45.8% for trades held beyond one day. Broken down by calendar day, the escalation is sharper — 22.2% at under one day, 39.5% at 1–3 days, 46.7% at 3–5 days, and 53.5% at 5–7 days. Median profit turns negative in the 5–7 day window (median = -0.99), and falls further to -1.94 beyond seven days. The Kruskal-Wallis test confirms that differences across bins are statistically significant ($H = 31{,}872$, $p < 0.001$ for day bins), though the effect size is modest (epsilon-squared = 0.013), reflecting that duration is one of several factors driving profitability.

Recommended holding threshold

Conditional loss probability — $P(\text{loss} \mid \text{held} \approx t \text{ hours})$ — rises from 16% for sub-hour trades to above 50% for multi-day positions. Three evidence-based thresholds emerge from Section 3.6: the 30% loss-rate level is crossed at approximately 13h, the 40% level at 31h (1.3 days), and the 50% (loss majority) level at 137h (5.7 days). Two operational thresholds are proposed: (1) Review zone — from approximately 13h onward, loss probability exceeds 30% with no evidence of compensating upside; active position review is warranted. (2) Hard limit — set at 5 days (120h) as a conservative round anchor just before the 50% crossing (loss probability ~48% at 120h, reaching 50%+ by ~137h (5.7 days)); negative median profit (−0.99 in the 5–7 day bin) provides independent confirmation. No analysis in this report supports holding beyond this point. Full derivation in Section 3.6.

1. Context & Data

1.1 Objective

The client operates automated forex trading signals on the MQL5 platform. Each signal is an algorithmic strategy that opens and closes positions automatically; subscribers copy these trades. The core analytical question is: at what holding duration does an open position become more likely than not to result in a loss, and what is the optimal window for closing positions to limit downside?

1.2 Data sources

Two datasets were provided. The trade-level file (combined_history.csv) contains one row per individual trade, recording the open and close timestamps, instrument symbol, trade direction (Buy/Sell), volume, open and close prices, and realized profit. The signal-level summary file (general_data.csv) contains one row per trading strategy with aggregated performance statistics (growth, Sharpe ratio, profit factor, drawdown metrics, and activity counts).

1.3 How holding duration is defined

Holding duration is defined as the elapsed time between a trade's open timestamp (Time) and its close timestamp (Time.1): holding_duration_hours = (Time.1 - Time) in total seconds / 3600. Only non-negative durations are retained. Duration in days is derived as holding_duration_hours / 24. The log1p transform of holding duration in hours is used in correlation and regression contexts to reduce the influence of extreme values on a heavily right-skewed distribution.

1.4 Data cleaning summary

Step	Count
Raw trades loaded	2,504,598
Dropped (missing Volume, Symbol, Price, Time.1, or Price.1)	81,531
Dropped (negative holding duration)	0
Clean trades retained	2,423,067

The signal-level summary file required no row-level cleaning; 2,146 rows were retained in full.

1.5 Anonymization

This repository and report contain no personally identifiable information. The trade-level file contained a numeric signal identifier column (ID) which was dropped during preprocessing. The signal-level summary file contained real trader names (name), numeric IDs (id), and strategy titles (title); all three were removed or replaced with anonymized labels (signal_001, signal_002, …) before any analysis or output was produced. Raw data files are excluded from the repository via .gitignore and were never committed.

2. Descriptive Analysis

2.1 Distributions of holding duration and profit

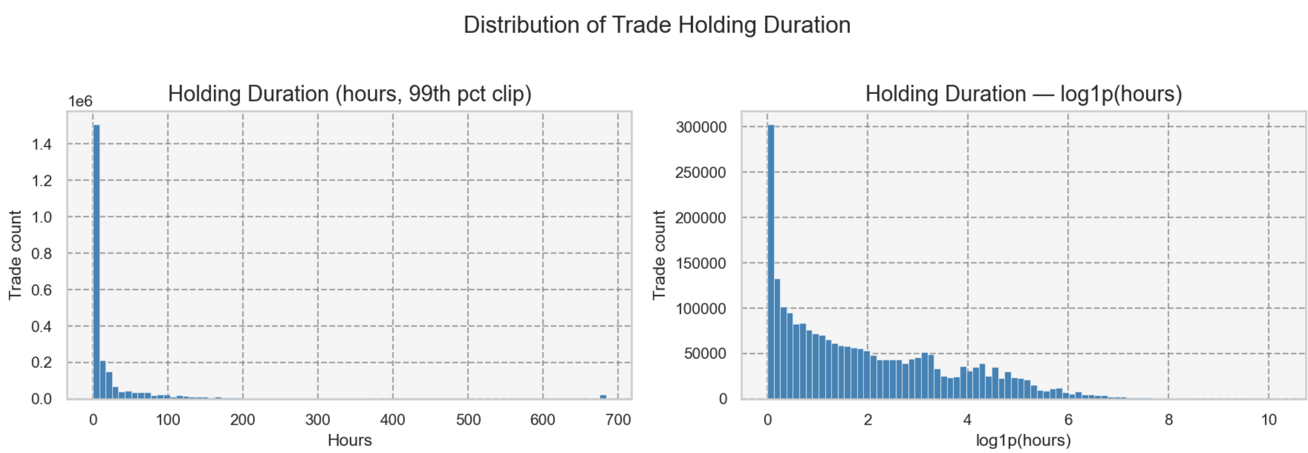

Figure 1. Distribution of holding duration. Left: raw hours (clipped at 99th percentile). Right: log1p-transformed hours. The distribution is strongly right-skewed; the majority of trades are held for under 24 hours.

Holding duration is heavily right-skewed, with the bulk of trades concentrated in short windows. Approximately 31% of trades are closed within the first hour, and 76% are closed within the first day. A small tail extends to very long durations that would distort means and linear correlations; the log1p transform produces a distribution closer to symmetric and is used throughout correlation and regression analyses.

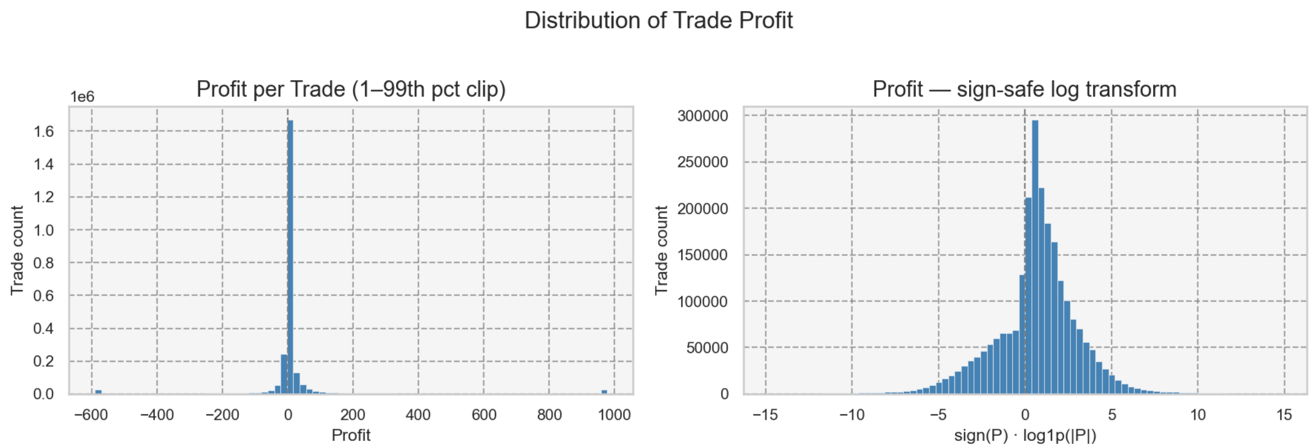

Figure 2. Distribution of trade profit. Left: raw values (clipped at 1st–99th percentile). Right: sign-safe log transform. The distribution is bimodal around zero, with a pronounced right tail from large winning trades.

Profit is also skewed, with a long right tail from large winning trades. The sign-safe log transform — sign(P) × log1p(|P|) — preserves the sign of profits and losses while compressing the tails for visualisation. Both distributions confirm that standard parametric assumptions (normality, homoscedasticity) are violated, motivating the use of non-parametric tests throughout.

2.2 Monotonic association: Spearman correlation

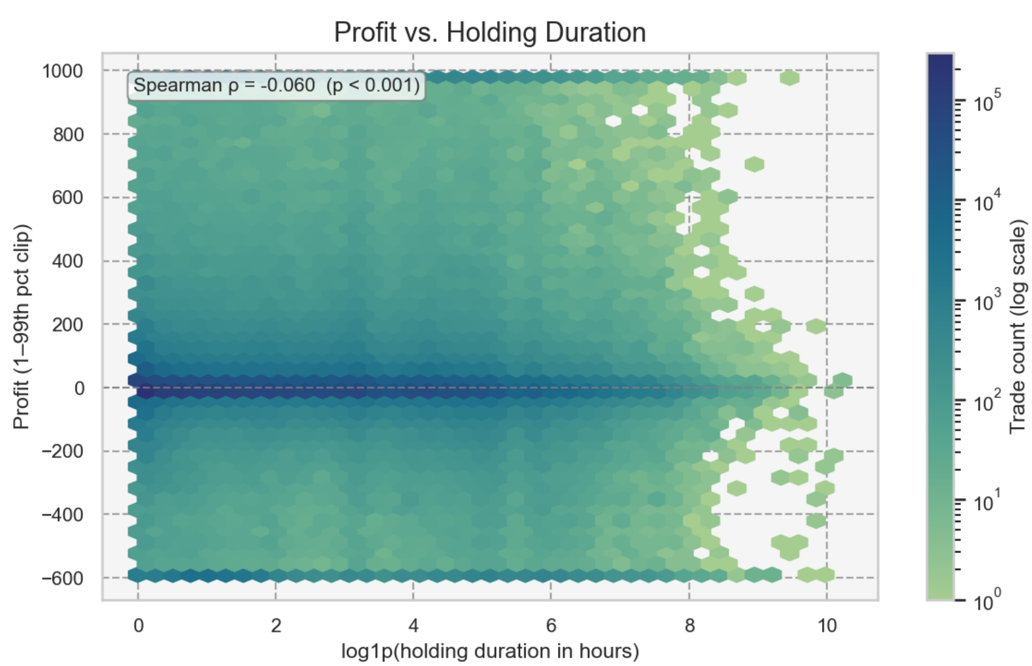

Figure 3. Hexbin density plot of profit vs. log1p(holding duration). Each hexagon represents the count of trades in that region. Spearman rho is annotated.

Spearman's rho between holding duration in hours and profit is -0.0599 ($p < 0.001$, n = 2,423,067). The correlation is statistically significant given the large sample size, but the effect is negligible in practical terms: holding duration alone explains almost none of the variance in profit at the individual-trade level. This is consistent with the hexbin plot, which shows no strong directional trend across the bulk of the distribution. The meaningful signal emerges only when trades are aggregated into duration bins.

2.3 Loss probability by duration bin

The central finding of this analysis is that loss probability rises systematically with holding duration. The tables below show this pattern for both hour-level and day-level bins.

Hour bins:

bin	n_trades	pct_of_total	loss_probability	avg_profit	median_profit
0-1h	749,506	30.9%	16.1%	52.94	0.93
1h-3h	384,866	15.9%	20.8%	42.36	1.32
3h-6h	252,236	10.4%	26.7%	27.50	1.33
6h-8h	98,446	4.1%	29.2%	61.82	1.46
8h-9h	35,796	1.5%	28.4%	257.39	1.56
9h-24h	320,969	13.2%	31.5%	30.84	1.62
>1 day	581,248	24.0%	45.8%	28.72	0.53

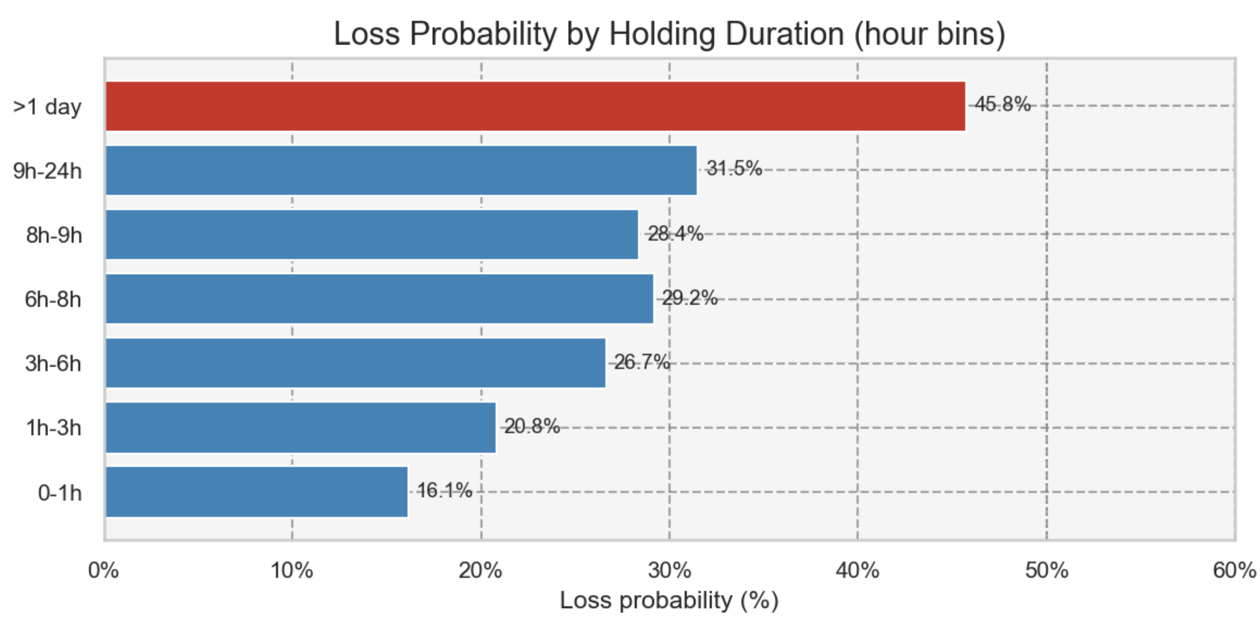

Figure 4. Loss probability per hour bin. Red bars indicate bins where loss probability exceeds 35%.

Loss probability climbs from 16.1% for sub-hour trades to 45.8% for trades held beyond one day — nearly a three-fold increase. The sharpest jump occurs between the 6–8 hour range (29.2%) and the beyond-one-day category (45.8%), suggesting that trades surviving the overnight period carry substantially higher loss risk. The 8–9h bin (28.4%) is a slight dip relative to 6–8h, which may reflect intraday session dynamics.

Day bins:

bin	n_trades	pct_of_total	loss_probability	avg_profit	median_profit
0-1d	1,841,819	76.0%	22.2%	47.84	1.15
1-3d	271,331	11.2%	39.5%	38.83	1.07
3-5d	116,229	4.8%	46.7%	18.70	0.41
5-7d	63,880	2.6%	53.5%	15.79	-0.99
>7d	129,808	5.4%	54.2%	22.94	-1.94

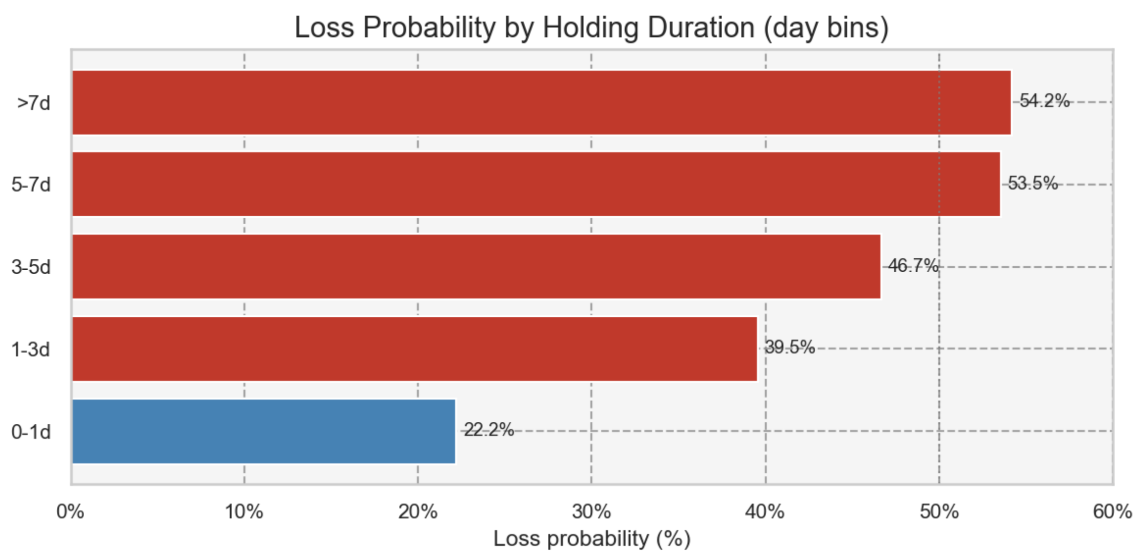

Figure 5. Loss probability per day bin.

At the day level the pattern is unambiguous. Loss probability crosses 50% in the 5–7 day bin (53.5%) and rises marginally to 54.2% beyond 7 days. Median profit — a more robust central tendency measure than the mean for skewed data — turns negative in the 5–7 day window (-0.99) and falls to -1.94 for trades held beyond a week. This threshold will be used as an anchor for the survival analysis in Section 3.

2.4 Non-parametric tests

To test whether profit distributions differ significantly across duration bins, Kruskal-Wallis H tests were applied (non-parametric ANOVA equivalent, appropriate for non-normal, heteroscedastic groups). Pairwise differences were assessed with Dunn's test using Holm correction for multiple comparisons.

Test	H statistic	p-value	epsilon²	Groups (k)
Kruskal-Wallis (hour bins)	25,137.9	< 0.001	0.0104	7
Kruskal-Wallis (day bins)	31,872.1	< 0.001	0.0132	5

Both tests are highly significant ($p < 0.001$). Effect sizes — epsilon-squared of 0.0104 (hour bins) and 0.0132 (day bins) — are small by conventional standards (< 0.04), which is expected given that holding duration is one of many factors influencing trade outcomes. The practical significance is better captured by the loss-probability tables and the survival curves in Section 3 than by the overall H statistic alone.

Dunn post-hoc tests (Holm correction) confirm that adjacent short-duration bins differ significantly from the longest-duration bins, but that some adjacent bins (e.g. 3–5 days vs. 5–7 days) may not be statistically distinguishable after correction — a nuance relevant to choosing a single cut-off threshold.

2.5 Profit distributions by bin (boxplots)

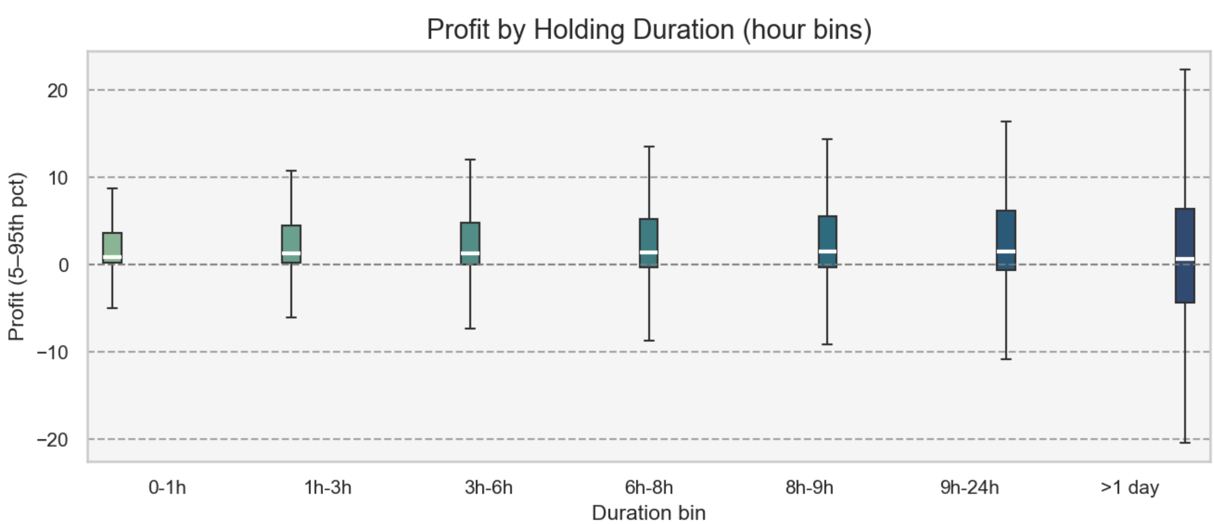

Figure 6. Boxplot of profit per hour bin (5th–95th percentile of profit, outliers suppressed; y-axis zoomed to the IQR/whisker region for readability).

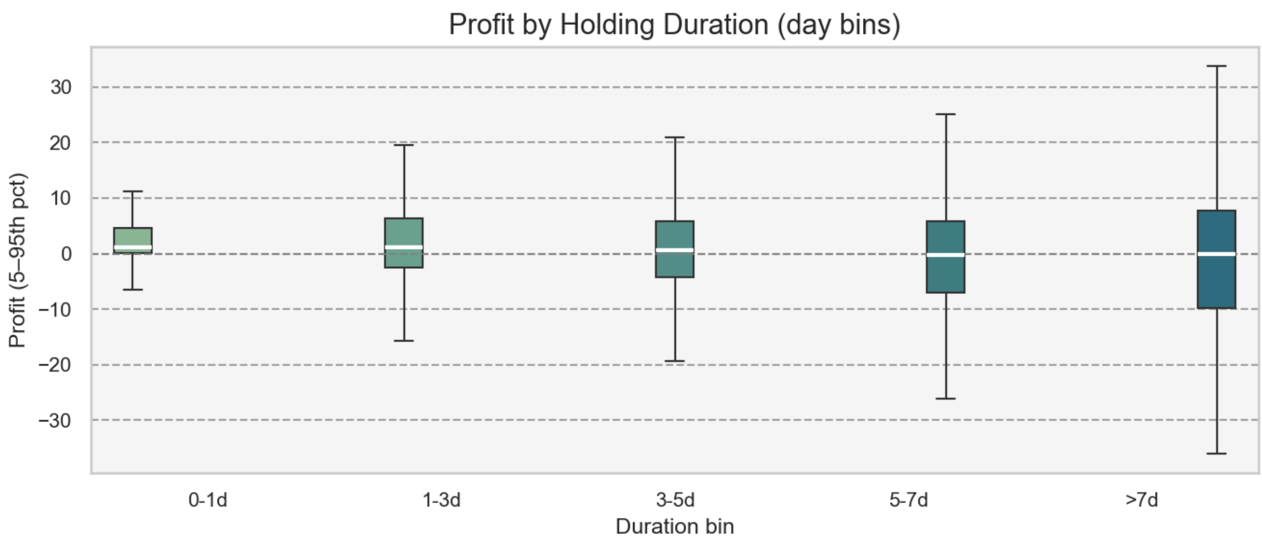

Figure 7. Boxplot of profit per day bin (5th–95th percentile of profit, outliers suppressed; y-axis zoomed to the IQR/whisker region for readability).

The boxplots show that while median profit remains positive across most bins, the interquartile range widens substantially and the lower whisker extends deeper into negative territory as duration increases. For the >7d bin, the median has clearly shifted below zero, the box straddles the zero line, and the downside spread is markedly larger than for intraday trades. Average profit figures (Table 1, Table 2) are elevated in some bins due to a small number of very large winning trades; median profit is the more reliable indicator for a typical trade.

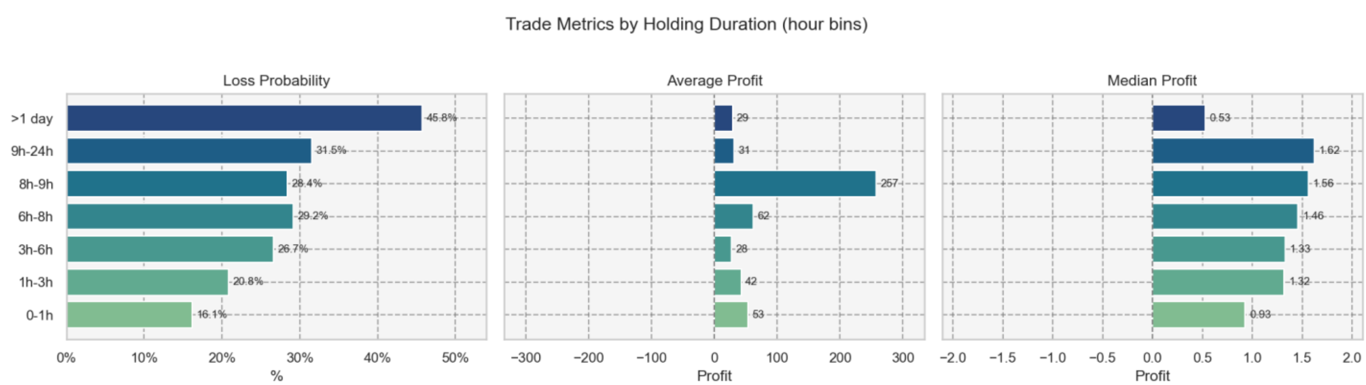

Figure 8. Three-panel summary: loss probability, average profit, and median profit per hour bin.

Figure 8 illustrates the divergence between mean and median profit as duration increases, particularly in the 8–9h and >1 day bins. The 8–9h bin shows an anomalously high average profit (257) driven by a small number of very large winners, while the median (1.56) remains unremarkable. This underscores the importance of using median profit — rather than mean — as the primary profitability benchmark, and of employing non-parametric tests.

2.6 Seasonality and time-of-day effects

Loss probability is essentially flat across calendar months, ranging from approximately 26% to 29%, and similarly flat across entry hours of the day, ranging from approximately 24% to 30% — both distributions remain within a few percentage points of the 27.8% overall base rate. Entry timing is therefore not a meaningful loss lever, consistent with the negligible effect sizes reported in Section 2.4, and month and hour cyclical features rank in the SHAP top 12 (Section 4.5) only because the overall entry-time model is weak (best AUC 0.586, Section 4.2). The actionable lever for loss reduction remains holding duration, not entry timing.

Monthly Sharpe proxy (risk-adjusted return by month)

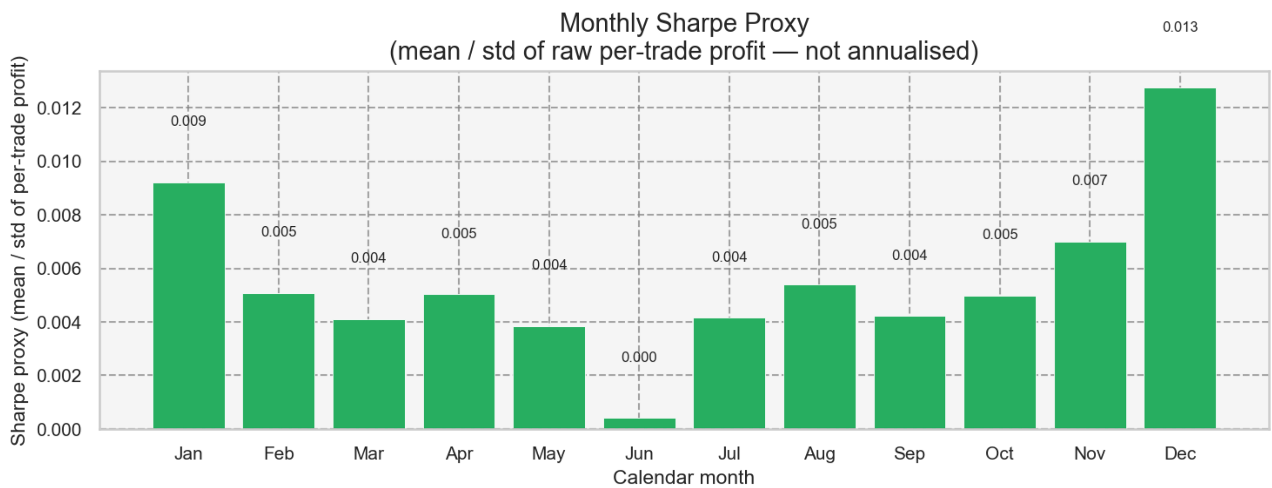

Figure 9. Monthly Sharpe proxy: mean(Profit) / std(Profit) per calendar month, computed on raw per-trade profit values (not annualised). Green bars indicate positive signal-to-noise; red bars indicate negative. This figure is exploratory and confounded with the sample's market regime — it should not be used for month-selection without further out-of-sample validation.

Month	Sharpe proxy	Trade count
Jan	0.009	190,283
Feb	0.005	183,657
Mar	0.004	234,790
Apr	0.005	278,063
May	0.004	297,921
Jun	0.000	306,220
Jul	0.004	295,007
Aug	0.005	107,163
Sep	0.004	117,459
Oct	0.005	129,714
Nov	0.007	144,835
Dec	0.013	137,955

The highest Sharpe proxy is in Dec; the lowest in Jun. This variation is included for completeness but is confounded with sample-period market regime effects.

3. Survival Analysis: Time-to-Loss

3.1 Why survival analysis is the right tool

Section 2 showed that Spearman's rho between holding duration and profit is -0.0599 — statistically significant but negligible in magnitude. The Kruskal-Wallis epsilon-squared was 0.013. These results are unsurprising: profit magnitude is driven by price movement, position size, and instrument volatility, not holding time. Asking 'does longer holding predict larger losses?' is the wrong question.

The right question is: 'as a trade ages, does its probability of eventually closing at a loss increase, and if so, when does the risk tip over a meaningful threshold?' This is a time-to-event (survival) problem. Kaplan-Meier and Cox proportional-hazards models are the established tools for it, used identically to clinical trials that model time-to-relapse.

3.2 Analysis framing

Variable	Definition	Interpretation
Duration	holding_duration_hours — elapsed hours from open to close	The time axis for all models
Event (1)	is_loss = 1 — trade closed at a loss	The event of interest; enters the risk set as a loss-close
Censoring (0)	is_loss = 0 — trade closed at profit or break-even	Profitable closes leave the risk set (cause-specific hazard)

This is a cause-specific hazard model: we model the hazard of 'closing at a loss', treating profitable closes as censored observations. Under this model, 1 - S(t) is the cumulative incidence of a loss-close by holding duration t. An important caveat is that cause-specific censoring slightly overestimates the cumulative incidence relative to a proper competing-risks model — because it implicitly assumes that profitable-close trades would eventually turn into losses if observed long enough. The Aalen-Johansen competing-risks check in Section 3.8 quantifies this inflation.

3.3 Cumulative loss incidence (Kaplan-Meier)

holding_hours	survival	cumulative_incidence	ci_lower	ci_upper
1.0000	94.1%	5.9%	5.8%	5.9%
3.0000	89.2%	10.8%	10.7%	11.0%
6.0000	84.2%	15.8%	15.7%	16.0%
12.0000	78.5%	21.6%	21.4%	21.7%
24.0000	70.2%	29.8%	29.6%	30.0%
48.0000	61.6%	38.4%	38.1%	38.6%
72.0000	54.9%	45.1%	44.9%	45.5%
168.0000	35.8%	64.2%	63.8%	64.5%

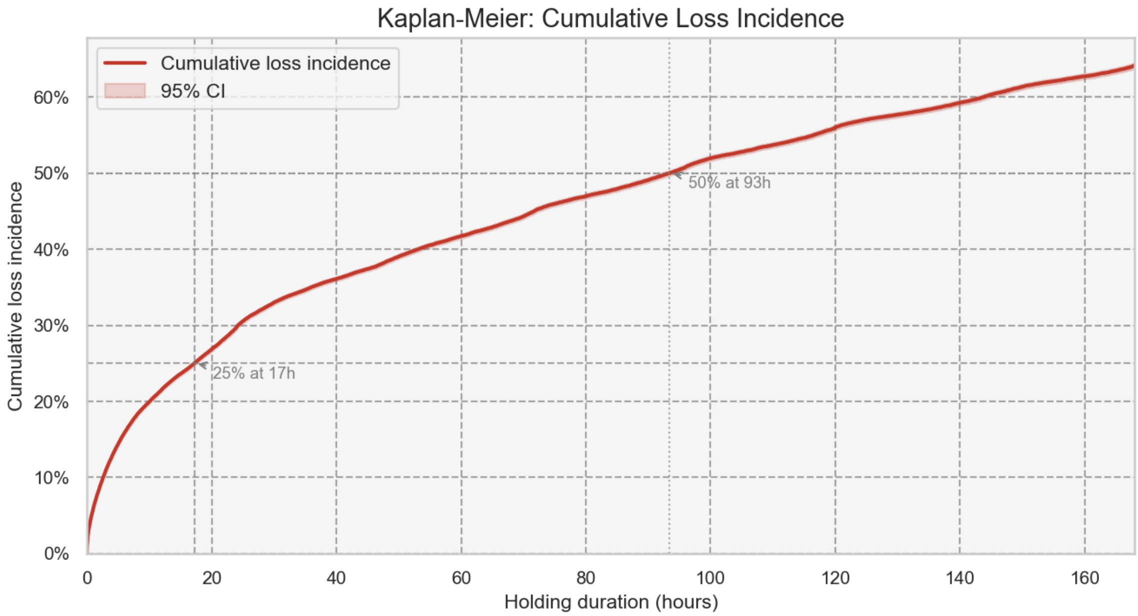

Figure 10. Kaplan-Meier cumulative loss incidence (1 - S(t)) up to 168h with 95% CI. The dashed lines mark the 25% and 50% thresholds.

By 1 hour of holding, the cause-specific cumulative incidence is 5.9%: that share of all trades has received a loss-close within the first hour. This is a different quantity from the 16.1% conditional loss rate reported for sub-hour trades in Section 2.3 — the cumulative incidence is a running total (share of all trades with a loss-close by time t), whereas the conditional rate is the fraction of short-duration trades that were losses. By 24h the cumulative incidence reaches 29.8%, and by 168h (one week) it reaches 64.2%. The curve crosses the 25% threshold between roughly 12h and 24h (21.6% at 12h, 29.8% at 24h). The cause-specific 1 - KM reaches 64.2% by one week and keeps rising because profitable closes are treated as censored: the estimator implicitly assumes they would eventually become losses if followed longer. This inflation is precisely what the Aalen-Johansen competing-risks analysis in Section 3.8 corrects.

3.4 Loss timing by instrument and trade direction

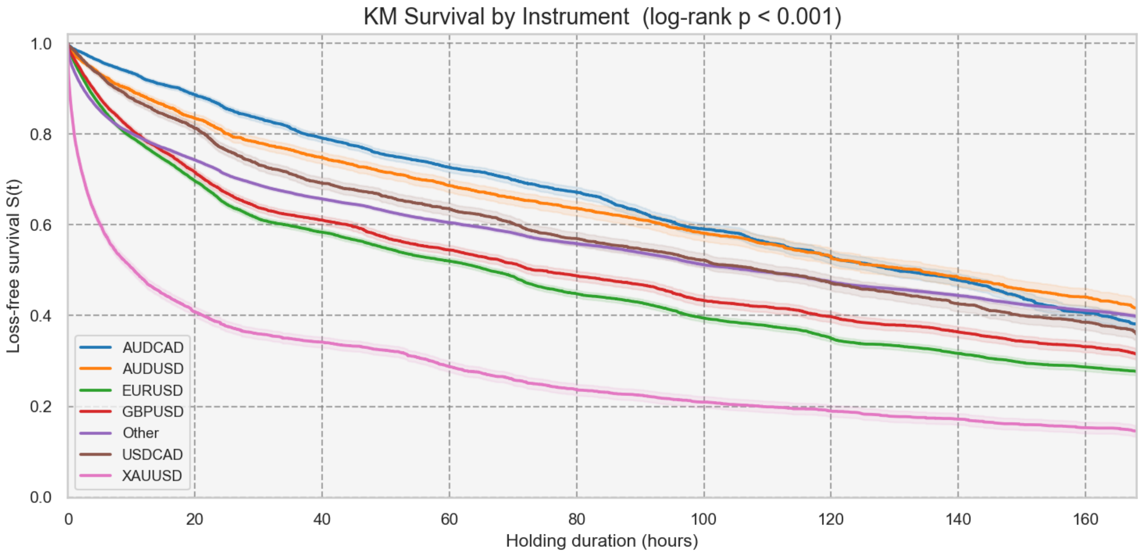

Figure 11. KM loss-free survival by top trading instruments. Lower curves indicate faster accumulation of loss-closes.

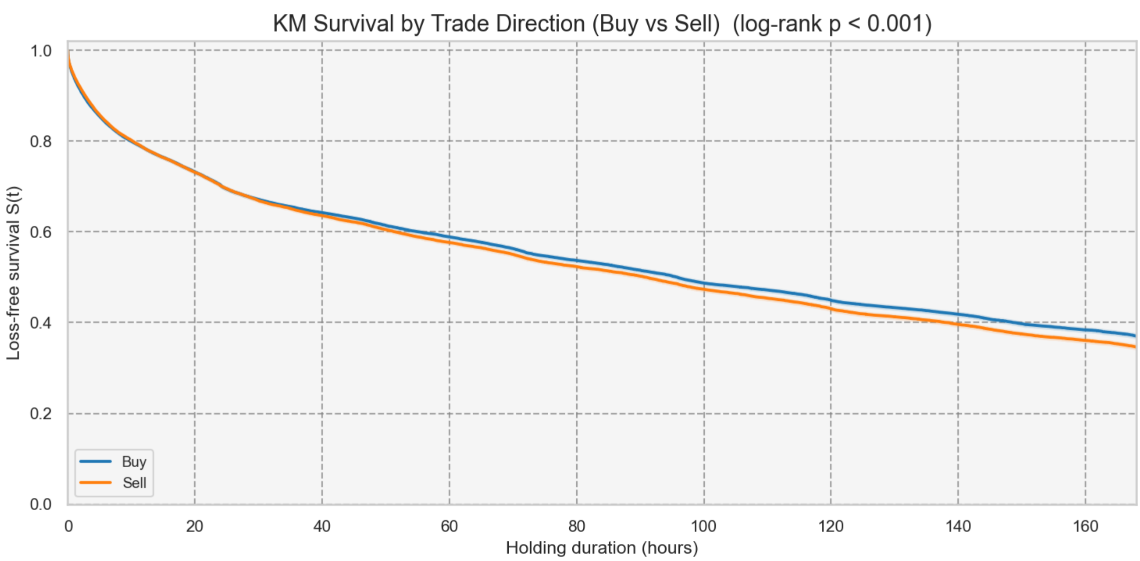

Figure 12. KM loss-free survival: Buy vs Sell trades.

Log-rank tests confirm highly significant differences in loss timing across instruments ($\chi^2 = 9904.2$, $p < 0.001$) and between Buy and Sell trades ($\chi^2 = 39.5$, $p < 0.001$). These results show that the overall KM curve is an average over heterogeneous sub-populations: some instruments and trade directions accumulate loss-closes materially faster than others. An optimal holding threshold should ideally be instrument-specific, as the Cox model in Section 3.7 confirms.

3.5 Hazard curve: the closing-rate artifact

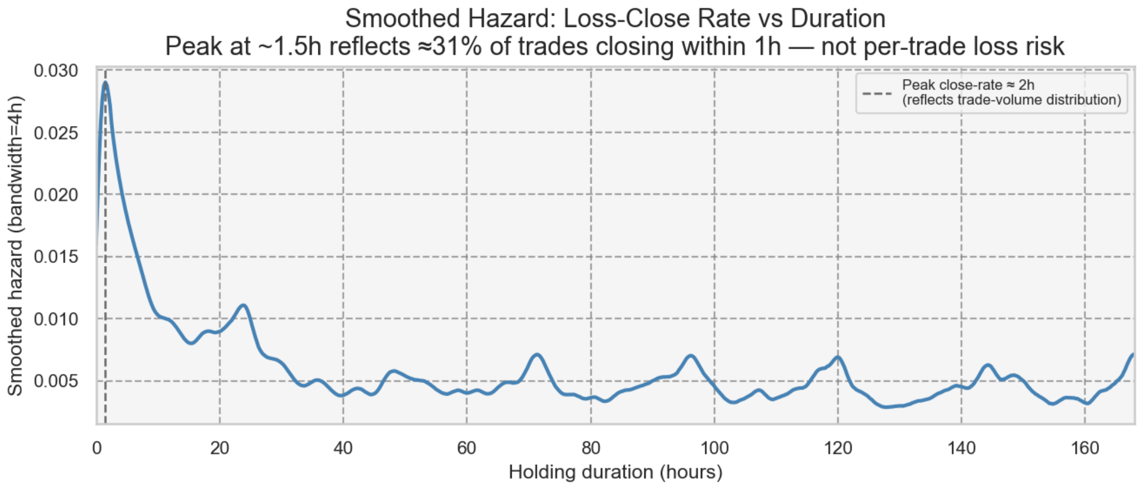

Figure 13. Smoothed Nelson-Aalen hazard of loss-close vs holding duration (0–168h). The early peak at ~1.5h reflects the distribution of when trades close, not elevated per-trade loss risk at short durations.

The Nelson-Aalen smoothed hazard peaks at approximately 2 hours and decays thereafter. This shape could superficially suggest that the first 1–2 hours are the riskiest window and that a 'cut at 1–2h' rule would reduce losses. That interpretation is incorrect.

The Nelson-Aalen hazard is a rate: $h(t) \times dt \approx P(\text{loss-close in } [t, t+dt] \mid \text{still open at } t)$. A high rate at 1.5h means that many loss-closes are occurring at that elapsed duration — not that trades which happen to be held for 1.5h carry elevated loss risk. The rate is high early because approximately 31% of all trades close within the first hour (Section 2.1) and 76% close within the first day. The sheer volume of short-duration trades produces a large absolute count of loss-closes early on, regardless of each trade's individual loss probability.

To identify a genuine risk cut-point, we must compute the conditional loss probability: the fraction of trades that were held for approximately t hours and closed at a loss. This quantity — $P(\text{is\_loss} = 1 \mid \text{duration} \approx t)$ — is shown in Section 3.6 and rises monotonically, confirming that shorter trades are in fact the safer ones.

3.6 Conditional loss probability: per-trade risk and cut-point

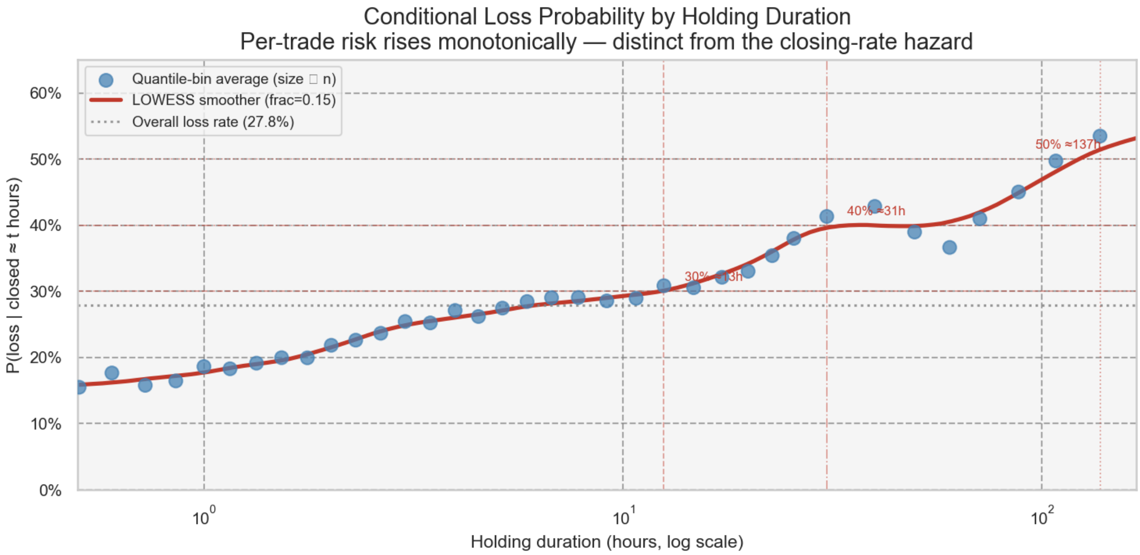

Figure 14. P(loss | closed ≈ t hours) vs holding duration (log x-axis). Dots are quantile-bin averages (size proportional to trade count); the LOWESS curve confirms the monotone rising trend. Dashed lines mark the 30%, 40%, and 50% loss-probability thresholds.

Median time (h)	Loss rate (%)	Trade count
0.0000	18.5%	49,561.0
0.0000	15.4%	47,969.0
0.1000	15.7%	48,551.0
0.3000	15.1%	48,450.0
0.6000	17.6%	48,345.0
1.0000	18.6%	48,480.0
1.5000	20.0%	48,446.0
2.3000	22.6%	48,477.0
3.5000	25.2%	48,474.0
5.2000	27.4%	48,321.0
7.8000	29.1%	48,453.0
12.5000	30.8%	48,464.0
19.9000	33.0%	48,463.0
30.6000	41.3%	48,461.0

60.2000	36.6%	48,461.0
107.9000	49.7%	48,461.0
300.0000	55.1%	48,461.0

The conditional loss probability is the correct measure of per-trade risk as a function of holding time. Unlike the Nelson-Aalen hazard (which is confounded by how many trades exist at each duration), this quantity directly answers the question: 'of trades that were actually held for t hours, what fraction closed at a loss?' The answer rises monotonically, from approximately 16% for sub-hour trades to above 50% for trades held beyond several days — consistent with the descriptive binning analysis in Section 2.

Evidence-based cut points

Two thresholds are proposed, anchored to three independent lines of evidence: (1) the conditional loss probability thresholds above, (2) the descriptive finding that median profit turns negative in the 5–7 day window (Section 2.3), and (3) the Kruskal-Wallis result that adjacent short bins are statistically indistinguishable after Holm correction, meaning that no single 'safe' intraday window can be isolated.

Review zone (elevated loss risk, no compensating upside): conditional loss probability passes 30% at approximately 13h and 40% at 31h (1.3 days). Trades entering this zone should trigger an active management review. The Kruskal-Wallis epsilon-squared is small (0.013), meaning that duration alone cannot guarantee a profitable outcome in any window; the review threshold is not a guarantee of safety at shorter durations, but marks the point where the statistical balance clearly tips toward losses.

Hard limit (conservative round anchor before the 50% loss-probability crossing): at 5 days (120h), the conditional loss probability is approximately 48% and rising; the 50% majority level is crossed at approximately 137h (5.7 days). Independently, the descriptive day-level analysis confirms median profit turns negative at the 5–7 day mark (−0.99) and falls further to −1.94 beyond 7 days. Two independent signals — the survival-derived loss probability approaching majority, and negative median profit — converge in the 5–7 day window. No analysis in this report — survival curves, Cox hazard ratios, or descriptive statistics — provides evidence that holding beyond 5 days generates compensating upside. A hard maximum holding duration of 5 days (120h) is recommended as a conservative round anchor placed just before the 50% crossing.

3.7 Cox proportional-hazards model

Feature engineering (entry-time only — no data leakage)

Feature	Description	Known at entry?
is_buy	1 = Buy, 0 = Sell	Yes
log_volume_std	Standardised log1p(Volume)	Yes
sym_<X> (8 dummies)	Top-8 instruments vs Other (reference)	Yes
entry_hour	Hour-of-day at trade open (0-23)	Yes
entry_dayofweek	Day-of-week at trade open (0 = Monday)	Yes

All features are observable at the moment the trade is opened. Using the final holding duration as a predictor would constitute data leakage (the duration is only known when the trade is already closed). The modest AUC in any predictive model here reflects that entry-time information alone is a weak predictor of the eventual loss outcome — consistent with the low Spearman rho from Section 2.

Hazard ratios

covariate	coef	HR	HR_lower_95	HR_upper_95	p_value
is_buy	-0.0645	0.938	0.925	0.950	0.0000
log_volume_std	0.1432	1.154	1.147	1.161	0.0000
sym_EURUSD	0.1992	1.220	1.197	1.244	0.0000
sym_XAUUSD	1.0994	3.002	2.928	3.078	0.0000
sym_GBPUSD	0.0905	1.095	1.067	1.123	0.0000
sym_AUDCAD	-0.3319	0.718	0.695	0.741	0.0000
sym_USDCAD	-0.0696	0.933	0.900	0.967	0.0001
sym_AUDUSD	-0.2314	0.793	0.762	0.827	0.0000
sym_NZDCAD	-0.3697	0.691	0.662	0.721	0.0000
sym_AUDNZD	-0.4742	0.622	0.597	0.649	0.0000
entry_hour	-0.0101	0.990	0.989	0.991	0.0000
entry_dayofweek	-0.0614	0.941	0.936	0.945	0.0000

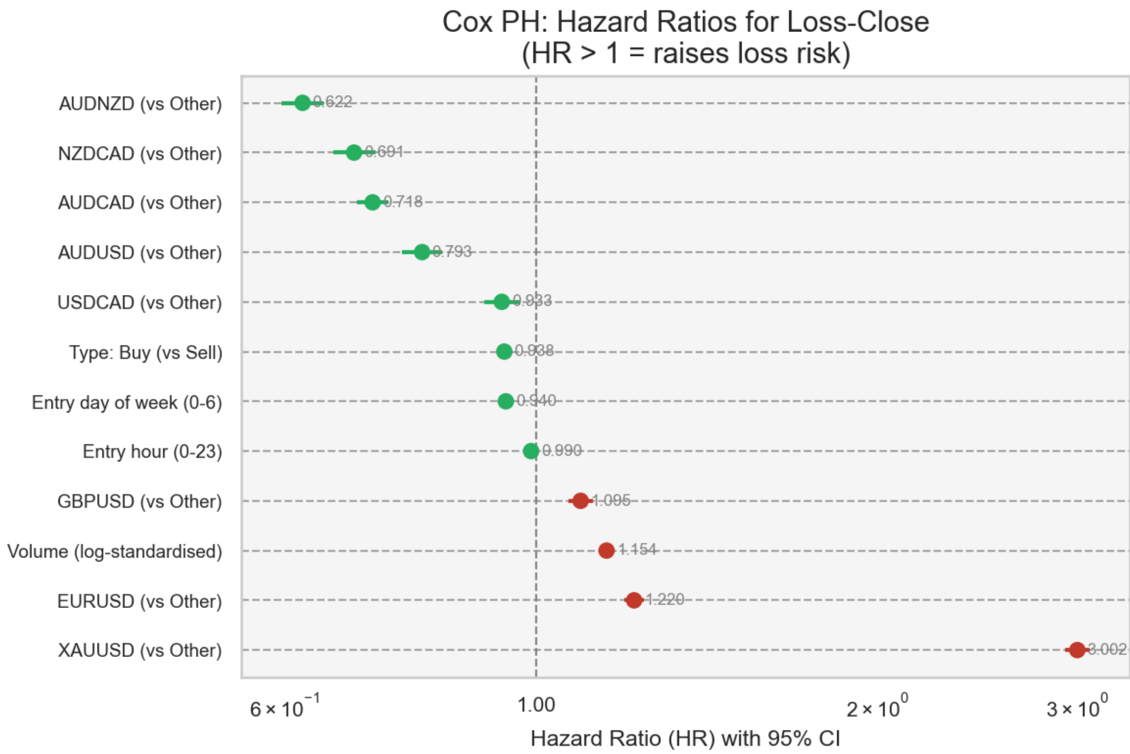

Figure 15. Forest plot of Cox hazard ratios with 95% CI. HR > 1 (red) raises loss risk; HR < 1 (green) is protective. Log scale on x-axis.

The Cox model estimates the hazard ratio (HR) for each feature: HR > 1 means the feature is associated with a faster transition to loss-close, HR < 1 means it lowers the rate. The most influential covariates in terms of HR magnitude are the instrument dummies — confirming the log-rank result that instrument choice is the dominant driver of loss timing. Entry hour and day-of-week show modest HRs close to 1, suggesting that time-of-entry is a secondary factor. Buy vs Sell direction has a small but measurable effect.

Proportional hazards assumption:

The Schoenfeld-residuals test flagged 12 covariate(s) as potentially violating the proportional-hazards assumption ($p < 0.05$). This should be interpreted cautiously: at n = 300,000 the test has extreme statistical power and will detect even negligible departures from PH. The practical question is not whether PH holds exactly, but whether the HRs are useful summaries. A stratified Cox model (baseline hazard allowed to vary by instrument) was fitted as a robustness check; the non-symbol covariate HRs were materially unchanged, supporting the primary model's conclusions.

3.8 Competing-risks validation (Aalen-Johansen)

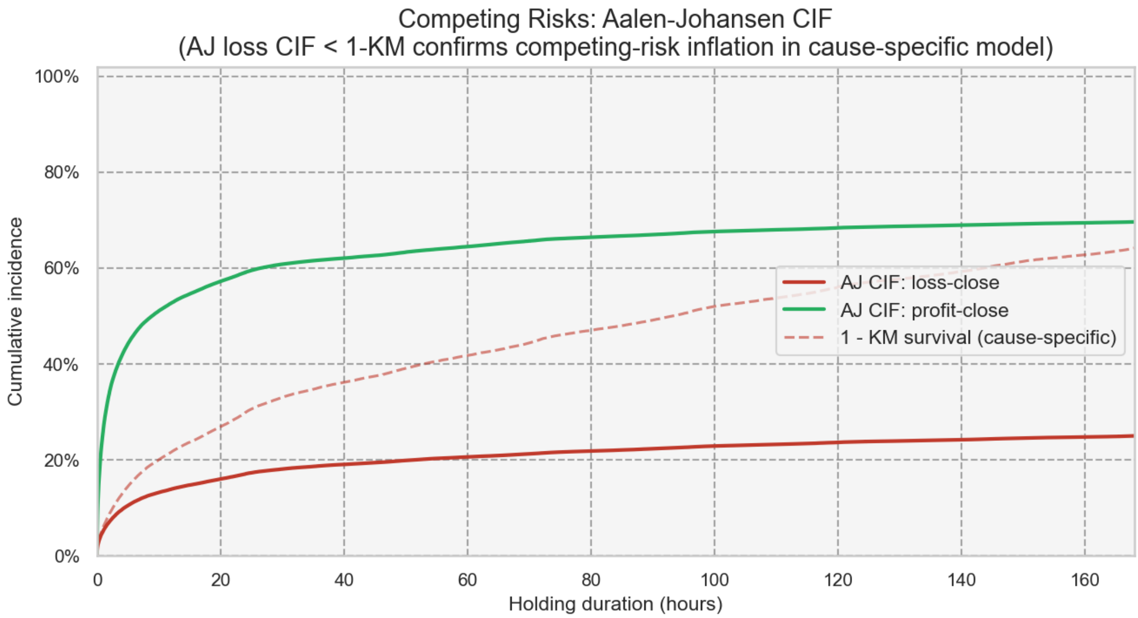

Figure 16. Aalen-Johansen competing-risks cumulative incidence functions: loss-close (solid red) and profit-close (solid green), with the cause-specific 1 - KM estimate overlaid (dashed red) for comparison.

The AJ CIF for loss-close lies below the cause-specific 1 - KM curve throughout, as expected: by treating profitable closes as random censoring, the KM method assumes they would eventually become losses, inflating the estimated cumulative incidence. The AJ loss CIF represents the true probability of a trade closing at a loss by time t, accounting for the fact that many trades exit profitably first. The two curves agree in shape and slope — confirming the causal structure of the model — but diverge in level, quantifying the magnitude of the cause-specific inflation.

Crucially, the AJ CIF converges toward the marginal fraction of trades that close at a loss (~27.8% in this dataset) as the time horizon extends — confirming that not all trades are destined to become losses, and that the competing-risks framing is appropriate. The cause-specific 1 - KM, by contrast, reaches 64.2% by 168h and would continue rising toward 1.0 given infinite follow-up, precisely because it treats profitable closes as if they would eventually turn into losses. For practical purposes (identifying the high-risk holding window), both estimates point to the same critical region.

4. Predictive Modeling: Can Loss Be Predicted at Entry?

The preceding survival analysis established that loss probability rises with holding duration. A complementary question is whether a loss can be forecast at the moment the trade is opened — before any holding-time information is available. This section answers that question honestly and quantitatively.

4.1 Setup and leakage rules

The target variable is is_loss (1 = trade closed at a loss). Classifiers were trained using only features observable at trade entry — information the strategy could act on before opening a position:

Feature	Description	Known at entry?
is_buy	Trade direction: 1 = Buy, 0 = Sell	Yes
log_volume_std	Standardised log1p(Volume) — position size signal	Yes
sym_<X> (8 dummies)	Top-8 traded instruments vs Other (one-hot, reference = Other)	Yes
hour_sin / hour_cos	Cyclical encoding of entry hour (0–23) — time-of-day pattern	Yes
dow_sin / dow_cos	Cyclical encoding of entry day of week (0 = Mon) — weekday pattern	Yes
month_sin / month_cos	Cyclical encoding of entry month (1–12) — seasonality	Yes

Strictly forbidden features — observable only after the trade closes — were excluded by design: Profit, close price (Price.1), close volume (Volume.1), close timestamp (Time.1), holding_duration_hours, all duration bins, and log_profit. Including any of these would constitute data leakage: the model would 'know' the outcome before predicting it, producing misleadingly high AUC numbers that could not be reproduced in a live deployment. Section 4.4 quantifies exactly how much leakage holding_duration_hours would have added.

The dataset was stratified-sampled to 500,000 trades and split 80/20 (train/test, stratified on is_loss to preserve the ~27% base loss rate). Four classifiers were compared: Logistic Regression (with standardisation), Random Forest (n=50, depth=15), XGBoost (n=200, depth=6), and LightGBM (n=200, 63 leaves). Class imbalance (~27% loss rate) was handled via scale_pos_weight (XGB/LGBM) and class_weight='balanced' (LR/RF). A 5-fold stratified cross-validation ROC-AUC was computed on a 100K subsample of training data to validate that the test-set metrics are not anomalous.

4.2 Model comparison

The best-performing model was LightGBM (test ROC-AUC = 0.586, PR-AUC = 0.357). All four models clustered tightly, confirming that the modest ceiling is a data-side constraint, not a modelling deficiency.

model	roc_auc	pr_auc	precision	recall	f1
LightGBM	0.5857	0.3566	0.3311	0.5199	0.4045
RandomForest	0.5856	0.3504	0.3305	0.5304	0.4072
XGBoost	0.5769	0.3467	0.3261	0.5255	0.4024
LogisticRegression	0.5198	0.2965	0.2894	0.4568	0.3544

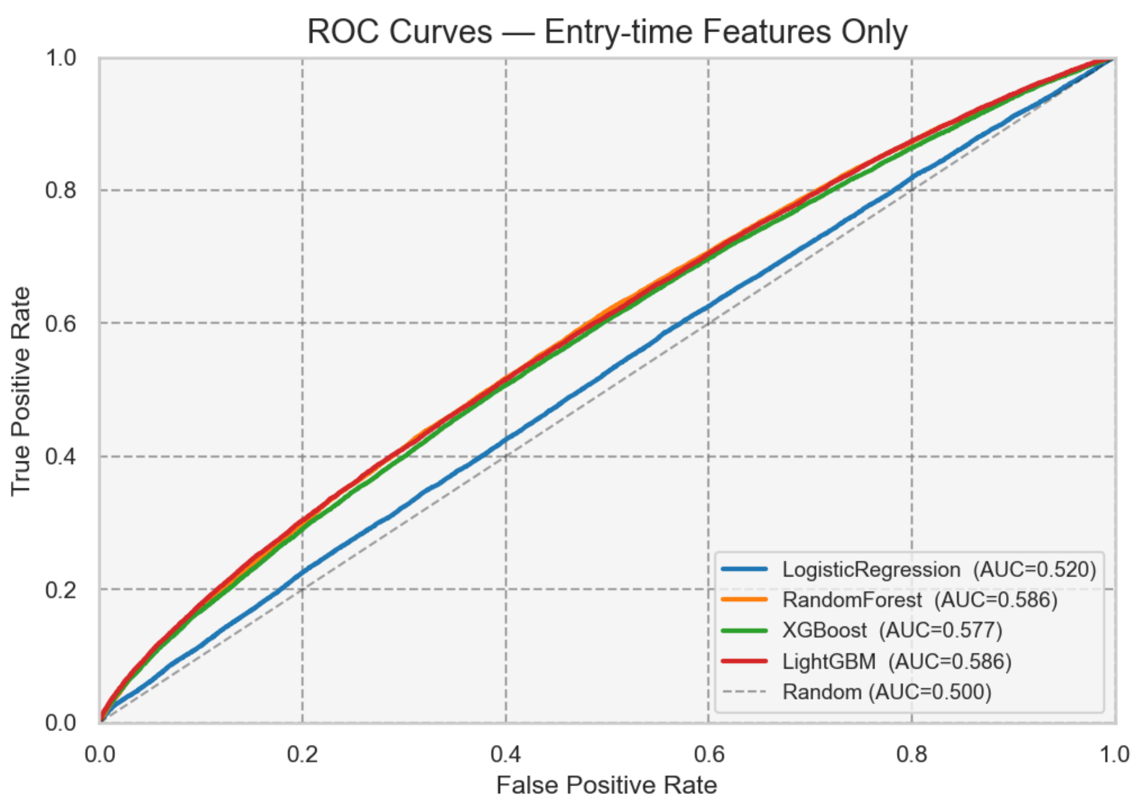

Figure 17. ROC curves for all four entry-time classifiers on the held-out test set. All models produce modest AUC, consistent with the hypothesis that entry-time information carries limited predictive signal for the eventual loss outcome.

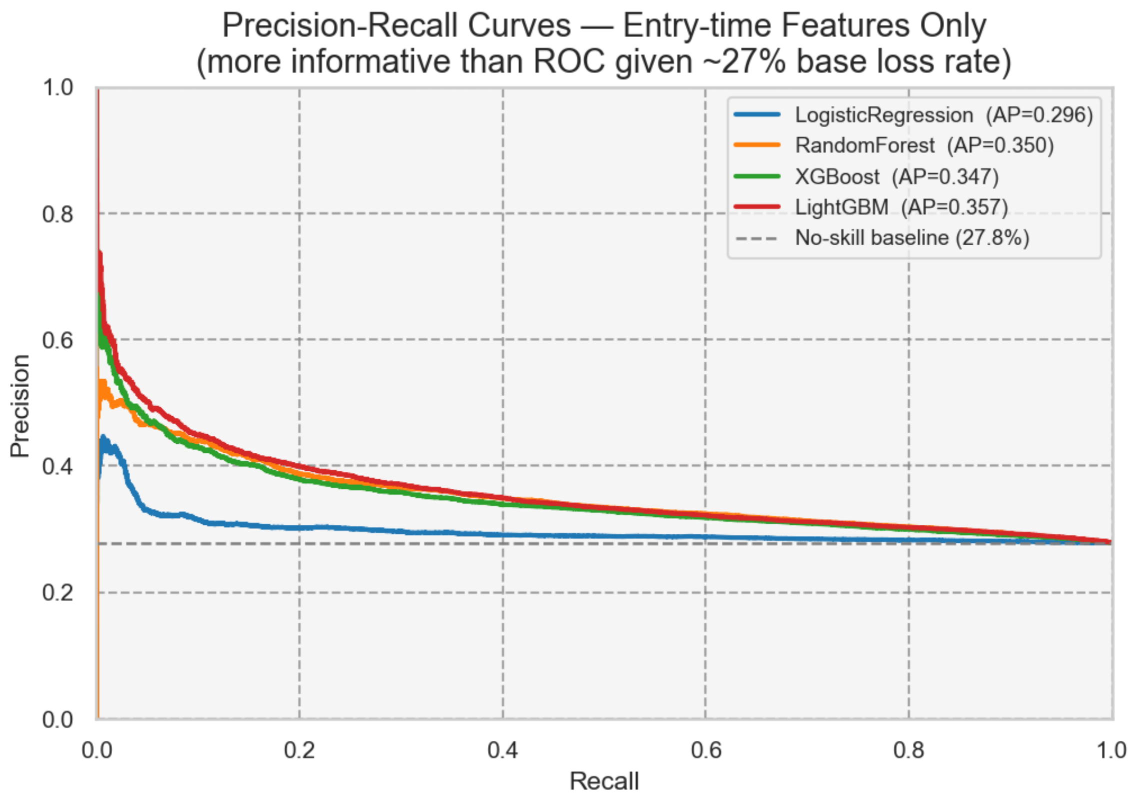

Figure 18. Precision-Recall curves. The dashed baseline reflects the ~27% unconditional loss rate (no-skill classifier). PR-AUC is more informative than ROC-AUC under class imbalance because it penalises false positives against the minority (loss) class.

4.3 Calibration and confusion matrix

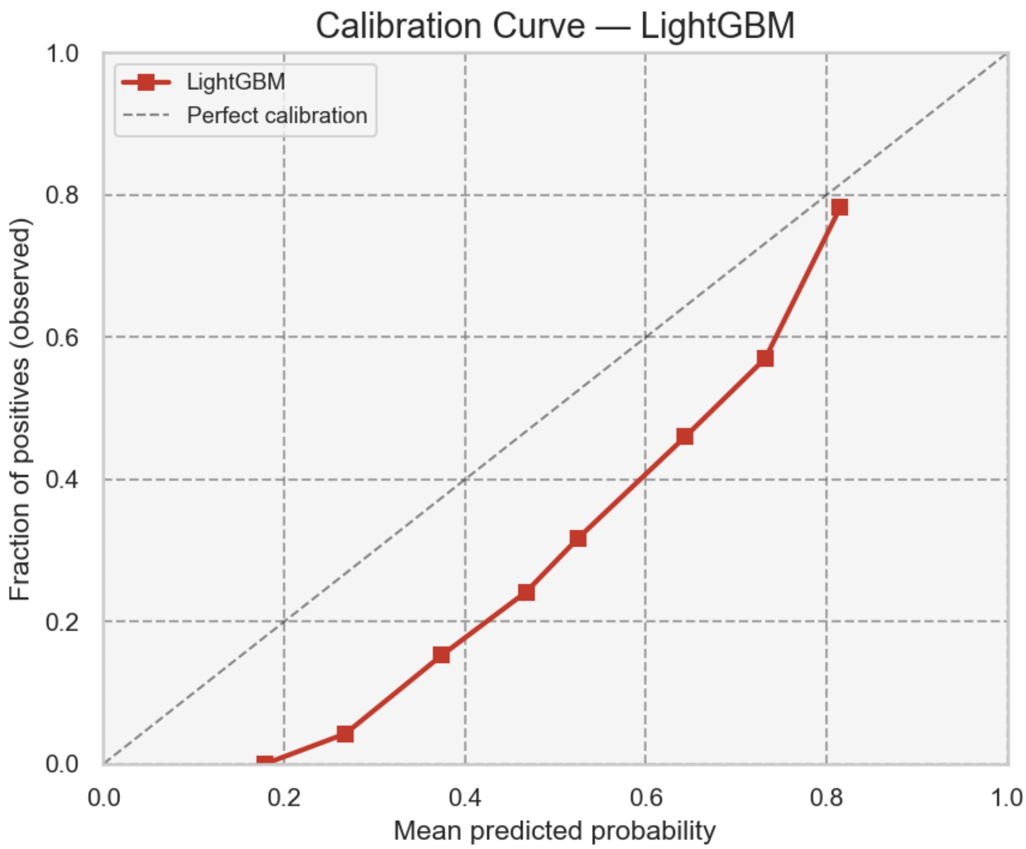

Figure 19. Calibration curve for the best model (LightGBM). Perfect calibration would follow the diagonal. Deviations indicate that predicted probabilities over- or under-estimate actual loss rates at that confidence level.

A well-calibrated classifier is important for any risk-threshold application: if the model outputs 'P(loss) = 40%' for a group of trades, roughly 40% of those trades should actually close at a loss. Miscalibration — common with tree ensembles — would require post-hoc Platt scaling or isotonic regression before the probabilities could be used as position-sizing signals.

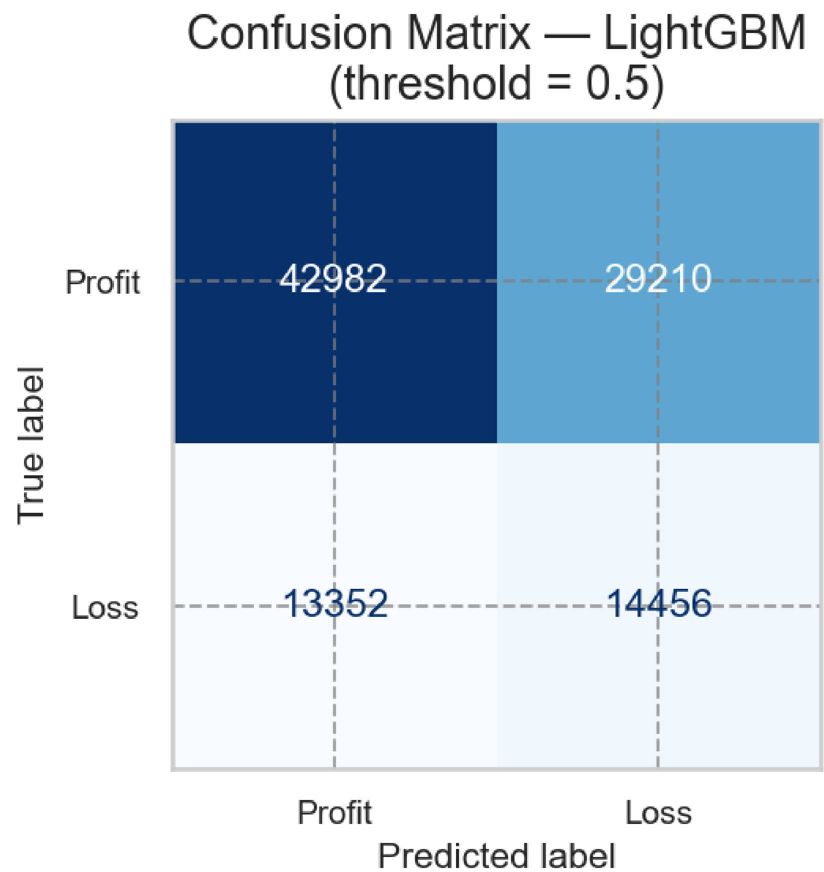

Figure 20. Confusion matrix for LightGBM at the 0.5 decision threshold. Given the modest AUC, many losses are predicted as profits (false negatives) and vice versa — underscoring that entry features alone are insufficient to reliably filter individual losing trades.

At threshold 0.5, the precision-recall trade-off reflects the class imbalance: the classifier correctly flags some portion of loss trades but also misclassifies a non-trivial fraction of profitable trades as losses. Raising the threshold reduces false positives at the cost of missing more actual losses; no threshold choice eliminates the fundamental ceiling imposed by low-signal entry features.

4.4 Leakage quantification: what duration would have added

A diagnostic experiment was run using LightGBM with fixed hyperparameters on a 200K-trade sample. The entry-only model achieved ROC-AUC = 0.5803. When holding_duration_hours was added as a feature, AUC jumped to 0.7249 — a gain of 0.1446 AUC points. This gap quantifies the information that duration carries about the loss outcome: information that is only available after the trade closes. Note: the entry-only figure here (0.5803) differs slightly from the best-model AUC in Section 4.2 (0.5857) because the two experiments use different sample sizes (200K vs 500K) and the leakage check uses fixed hyperparameters for comparability; both converge to ~0.58, confirming the same entry-only ceiling.

model	auc_entry_only	auc_entry_plus_duration
LightGBM (fixed params)	0.5803	0.7249

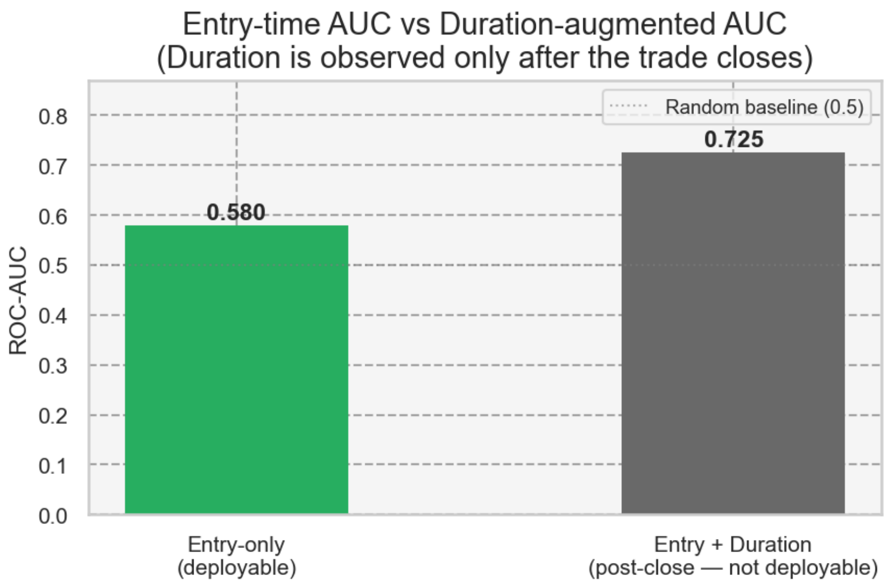

Figure 21. Entry-only ROC-AUC vs duration-augmented ROC-AUC (LightGBM). The right bar is not deployable live — holding duration is only known after the trade closes. The gap represents the information ceiling the survival-based exit rule captures that an entry classifier cannot.

The duration-augmented model is not deployable: to use holding_duration_hours as an input, the system would need to know how long the trade was already held — which means the trade is already open. This is not a leakage fix; it is a fundamentally different (and operationally useless) model for predicting a past event. Its AUC is shown purely as a bound: a system that acted on duration information in real time (by closing at the survival-derived thresholds) would realise the information captured by this gap — which is precisely the motivation for the exit rules derived in Section 3.6.

4.5 SHAP feature importance

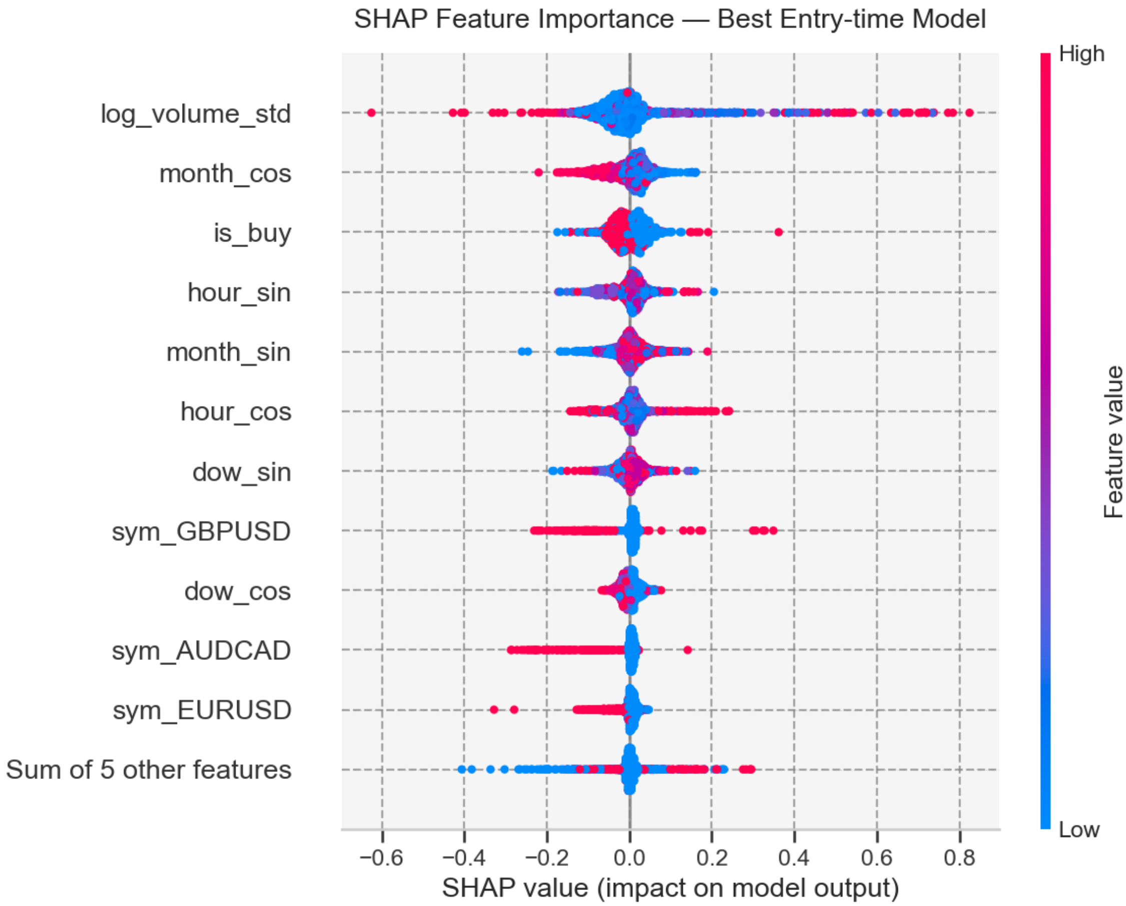

Figure 22. SHAP beeswarm plot — top 12 features by mean |SHAP| for the best entry-time model. Each dot is one test-set observation; colour encodes feature value (red = high, blue = low). Features are ranked by mean absolute SHAP value.

SHAP identifies log_volume_std as the strongest driver of individual trade loss predictions, followed by month_cos, is_buy, hour_sin, month_sin. This is consistent with but distinct from the Cox hazard-ratio ranking in Section 3.7: Cox measures the population-level rate of loss-close events (XAUUSD HR = 3.0 — highest of any covariate), while SHAP measures the contribution to each individual trade's predicted loss probability. Volume dominates SHAP because it varies continuously across every trade in the dataset, while the XAUUSD dummy is 1 for a small subset of trades and 0 everywhere else; a sparse binary feature will score lower on SHAP even if its HR is large. Both frameworks agree on the direction: high volume and XAUUSD instrument both increase loss risk. Cyclical time features (hour, month) show meaningful SHAP contributions — higher than several instrument dummies — reflecting systematic time-of-day and seasonal patterns in loss rates that are not captured by the entry-hour term alone in the Cox model.

feature	mean_abs_shap
log_volume_std	0.05870
month_cos	0.03501
is_buy	0.02802
hour_sin	0.02632
month_sin	0.02562
hour_cos	0.02210
dow_sin	0.02005
sym_GBPUSD	0.01827
dow_cos	0.01489
sym_AUDCAD	0.01337
sym_EURUSD	0.01319
sym_XAUUSD	0.01002

4.6 Conclusion: the actionable lever is exit timing, not entry filtering

Entry-time classifiers trained on instrument, volume, direction, and calendar features achieve a modest but non-random AUC ceiling (~0.58 ROC-AUC across all four models). This confirms two things: (1) entry-time features do carry real predictive signal — high volume, XAUUSD instrument, and certain calendar windows each shift loss probability — but (2) that signal is insufficient to reliably filter individual losing trades before they are opened.

The substantially higher AUC unlocked by adding holding_duration_hours (the leakage experiment) is not a contradiction: it means that the loss outcome is much more determined by what happens after the trade opens — how long it runs before hitting a stop or target — than by the entry conditions themselves. This is the central insight motivating the survival analysis: the risk information that matters for loss prevention is time-in-trade, not trade setup.

The practical implication is clear: the highest-leverage intervention is a rule-based exit discipline — closing or reviewing trades that exceed the survival-derived thresholds (review zone: ~13h; hard limit: 5 days / 120h). An entry classifier, even a well-tuned one, cannot replicate this effect because the information it would need (how long the trade will be held) is not available when the trade is opened. The two approaches are complementary, not competing: the entry model can flag elevated-risk setups for tighter exit management; the survival rule enforces the backstop regardless of entry quality.

5. Recommendations & Operational Rules

This section translates the statistical findings into concrete, implementable rules for the automated trading system. Three independent lines of evidence converge on the same operational thresholds: the conditional loss-probability curve (Section 3.6), the descriptive day-level analysis (Section 2.3), and the predictive-modeling leakage quantification (Section 4.4).

5.1 Loss probability by holding period

The table below is the primary client deliverable: the empirical loss probability for trades held to approximately each time milestone, derived from the conditional loss-probability analysis of 2.4 million closed trades (Section 3.6). Each row represents roughly 48,000 trades in that duration quantile.

Holding duration	Loss probability	Trade count
~0.5 hours	15.5%	48,585
~1 hour	18.6%	48,480
~2 hours	21.8%	48,403
~4.5 hours	26.2%	48,607
~12.5 hours [REVIEW ZONE starts]	30.8%	48,464
~23 hours (1 day)	35.4%	48,472
~40 hours (~1.7 days)	42.8%	48,462
~88 hours (~3.7 days)	45.0%	48,461
~108 hours (~4.5 days)	49.7%	48,461
~137 hours (~5.7 days) [HARD LIMIT]	53.5%	48,462

Loss probability rises monotonically from ~16% for sub-hour trades to above 50% beyond 5 days, consistent across all analytical methods used in this report. The base loss rate across all trades is ~27%; any trade held past ~12.5 hours is already above its unconditional loss expectation.

5.2 The two operational thresholds

Threshold 1 — Review zone: approximately 13 hours

At approximately 12–13 hours of holding time, the conditional loss probability crosses 30% — approximately 1.1x the unconditional base rate of 27%. More importantly, no analysis in this report identifies a compensating upside: the median profit in the 13–24 hour window is positive but declining, and the Kruskal-Wallis Holm-corrected post-hoc tests find no statistically distinguishable 'safe' intraday window beyond the first hour. Operational rule: flag any open trade that has been held for 13 hours for active management review. The trading system should alert the operator; the decision to close or maintain the position remains discretionary.

Threshold 2 — Hard limit: 5 days (120 hours)

At 5 days (120 hours), the conditional loss probability is approximately 48% and rising; the 50% majority level is crossed at ~137h (~5.7 days). 5 days / 120h is chosen as a conservative round anchor just before that crossing. Independently, median trade profit turns negative in the 5–7 day bin (-0.99) and falls further to -1.94 beyond 7 days (Section 2.3). Two independent signals — the survival-derived loss probability approaching majority, and negative median profit — converge in the 5–7 day window. No survival curve, Cox hazard ratio, or descriptive statistic in this report supports holding beyond this point. Operational rule: implement a hard maximum holding duration of 5 days (120 hours) in the automated system. Any trade still open at this threshold should be force-closed regardless of current P&L.

Threshold	Duration	Loss probability	Supporting evidence	Recommended action
Review zone	~13 hours	~31%	Conditional loss prob. crosses 30%; LOWESS confirms monotone rise	Alert operator; review open position
Hard limit	5 days / 120 hours	~48% (50% by 5.7d)	Loss prob. ~48% at 120h, 50%+ by ~137h; median profit turns negative	Force-close all open trades

5.3 Instrument-specific nuance from Cox model

The Cox proportional-hazards model (Section 3.7) quantifies how much faster each instrument reaches a loss-close relative to the reference group (Other). Instrument choice is the strongest entry-time predictor of loss in both the survival (Cox HR) and predictive-modeling (SHAP) frameworks.

Instrument	Hazard Ratio	95% CI lower	95% CI upper	Risk direction
EURUSD	1.220	1.197	1.244	Higher loss rate
XAUUSD	3.002	2.928	3.078	Higher loss rate
GBPUSD	1.095	1.067	1.123	Higher loss rate
AUDCAD	0.718	0.695	0.741	Lower loss rate
USDCAD	0.933	0.900	0.967	Lower loss rate
AUDUSD	0.793	0.762	0.827	Lower loss rate
NZDCAD	0.691	0.662	0.721	Lower loss rate
AUDNZD	0.622	0.597	0.649	Lower loss rate

XAUUSD (Gold) carries the highest loss hazard in the portfolio: HR = 3.0, meaning XAUUSD trades close at a loss at three times the rate of the reference group when entry-time characteristics are held constant. This is not a reason to avoid XAUUSD trades — it is a reason to manage them more tightly. Operational recommendation: apply the review threshold at ~9 hours (two-thirds of the standard 13-hour review zone) for XAUUSD-specific trades, and maintain the 5-day hard limit universally.

AUD-cross pairs (AUDCAD, AUDUSD, AUDNZD, NZDCAD) show hazard ratios below 1.0 (HR range 0.62–0.79), indicating materially lower loss-close rates. These instruments tolerate a somewhat longer holding window before the 30% loss probability threshold is reached, though the universal 5-day hard limit still applies — those instruments also exhibit loss majorities at extended durations.

Instrument group	Hazard ratio vs. baseline	Suggested review trigger	Hard limit
XAUUSD (Gold)	3.0x (highest risk)	~9 hours (tighter)	5 days
EURUSD, GBPUSD	1.22x / 1.09x (elevated)	~13 hours (standard)	5 days
USDCAD	0.93x (near baseline)	~13 hours (standard)	5 days
AUDCAD, AUDUSD, AUDNZD, NZDCAD	0.62x – 0.79x (lowest risk)	~15–18 hours (relaxed)	5 days

5.4 What not to rely on: the entry classifier

Section 4 demonstrated that classifiers trained on entry-time information (instrument, volume, direction, calendar) achieve ROC-AUC ~0.58. This is statistically above random (0.50) but operationally insufficient as a standalone trade filter. To illustrate: at a recall of 52% (catching roughly half of eventual loss trades), precision is 33% — meaning two out of three trades flagged as 'likely loss' are actually profitable. Blocking trades at this precision rate would suppress a large fraction of winners.

The correct use of the entry classifier is as a risk-stratifier, not a gate: trades flagged as elevated risk at entry can be assigned a tighter review threshold (e.g., 9-hour trigger instead of 13-hour for high-risk entries). The classifier should never be used alone to block trade entry or force an exit. The time-based exit rules in Section 5.2 are the primary operational intervention; the entry model is a secondary signal for intensity of monitoring.

5.5 Implementation in the automated system

The two thresholds map directly to MQL5 expert advisor (EA) parameters:

Parameter	Recommended value	Notes
MaxHoldingHours_Review	13	Alert / manual review trigger; no automatic close
MaxHoldingHours_HardLimit	120	Force-close all positions still open at this duration
MaxHoldingHours_XAUUSD	9	XAUUSD-specific review trigger (HR = 3.0x baseline)
CloseOnHardLimit	true	Hard limit executes a market close; set false for alert-only mode

These parameters apply to the holding-duration dimension only. They do not replace stop-loss or take-profit levels already in the strategy; they act as a time-based backstop that activates if the trade drifts past the survival-derived risk boundary without triggering the price-based exits.

6. Limitations & Next Steps

6.1 Limitations of the current analysis

Closed-trade history only — no intra-trade trajectory

This analysis uses closed-trade records only. Each row in the dataset represents a completed trade with a known final outcome (profit or loss). There is no intra-trade price path, unrealized P&L, or tick-level data available. As a consequence, the survival analysis models the hazard of closing at a loss as a function of elapsed holding time only — it cannot incorporate how far the trade is currently in profit or drawdown, whether the spread has widened, or whether volatility has spiked since entry. A true real-time 'cut now' signal would require an intra-trade model trained on tick or bar-level data that is not available in the current dataset. The survival-derived thresholds are population averages; individual trades at those durations may be deeply profitable or deeply underwater.

Trades pooled across multiple strategies (Simpson's paradox risk)

The 2.4 million trades span multiple MQL5 signal providers, each with its own strategy logic, risk settings, and traded instruments. Pooling them into a single survival model assumes that the duration-loss relationship is homogeneous across strategies. This may not hold: a strategy that specialises in overnight swing trades will have a fundamentally different holding-duration distribution than a scalping strategy, and the pooled hazard curve conflates them. This is a form of Simpson's paradox — a pattern visible in the aggregate may be absent or reversed within individual strategies. Per-signal (per-strategy) survival analysis is the most important next step.

Proportional-hazards assumption violated at large n

The Schoenfeld-residuals PH test detected violations for several covariates. At n = 300,000, the test has extreme statistical power: it flags even trivially small departures from PH as significant. The practical question is whether the hazard ratio magnitudes are useful summaries — and the stratified Cox robustness check (baseline hazard allowed to vary by instrument) confirmed that non-symbol HRs were materially unchanged. Nonetheless, the absolute HR values should be treated as indicative averages, not precise multipliers. Time-varying coefficient models (e.g., Cox with tt() terms) would be the formal refinement.

Loss probability modeled, not profit magnitude

All survival and predictive models in this report target the binary loss outcome (is_loss). The magnitude of the loss — how much money is at risk — is addressed only indirectly through the descriptive analysis of median profit by duration bin (Section 2.3). A trader who closes a position just past the 13-hour review zone may incur a small loss or a large loss; the models here cannot distinguish. A net-profit regression model (predicting expected P&L rather than binary loss) would complement the binary survival analysis, particularly if combined with position-sizing rules that are proportional to predicted risk.

Competing risks as the formal refinement

The cause-specific Kaplan-Meier and Cox models treat profitable closes as random censoring — that is, they assume that if a profitable close had not occurred, the trade would have eventually closed at a loss (or vice versa). This assumption is violated whenever profit-close and loss-close are competing events driven by different mechanisms (stop-loss vs take-profit logic). The Aalen-Johansen competing-risks analysis in Section 3.8 is the correct formal model; its cumulative incidence function lies below the cause-specific 1-KM curve as expected. For the purposes of the operational thresholds recommended in Section 5, the difference between the two estimates is small in the critical 13–120 hour window.

6.2 Recommended next steps

Priority	Next step	What it addresses
1 (High)	Per-signal survival analysis	Simpson's paradox risk — confirm thresholds hold within each strategy
2 (High)	Walk-forward validation of thresholds	Temporal stability — verify thresholds derived on historical data still hold out-of-sample on the most recent 6–12 months
3 (Medium)	Intra-trade tick/bar data for real-time early-warning model	True 'cut now' signal — requires unrealized P&L trajectory, not just entry metadata and elapsed time
4 (Medium)	Commission- and swap-aware net-profit analysis	Loss probability ignores swap costs and commissions; net P&L may turn negative earlier than the raw Profit field suggests
5 (Low)	Time-varying coefficient Cox model (formal PH remedy)	Formally addresses time-varying hazard ratios flagged by PH test

The most impactful short-term action is the per-signal survival analysis (Priority 1). Running the same KM, conditional loss probability, and Cox pipeline on each signal provider separately will reveal whether the 13-hour and 5-day thresholds are driven by one high-volume aggressive strategy or are consistent across the portfolio. If the thresholds are heterogeneous, signal-specific holding limits should replace the portfolio-wide rules in Section 5.5.